

Text and Characters

Text in String Arrays

When you are working with text, enclose sequences of characters in double quotes. You can assign text to a variable.

```
t = "Hello, world";
```

If the text includes double quotes, use two double quotes within the definition.

```
q = "Something ""quoted"" and something else."
```

```
q =
```

```
"Something "quoted" and something else."
```

t and q are arrays, like all MATLAB® variables. Their *class* or data type is string.

```
whos t
```

Name	Size	Bytes	Class	Attributes
t	1x1	174	string	

To add text to the end of a string, use the plus operator, +.

```
f = 71;
c = (f-32)/1.8;
tempText = "Temperature is " + c + "C"
```

```
tempText =
```

```
"Temperature is 21.6667C"
```

Similar to numeric arrays, string arrays can have multiple elements. Use the `strlength` function to find the length of each string within an array.

```
A = ["a", "bb", "ccc"; "dddd", "eeeeee", "fffffff"]
```

```
A =
```

```
2x3 string array
```

"a"	"bb"	"ccc"
"dddd"	"eeeeee"	"fffffff"

```
strlength(A)
```

```
ans =
```

1	2	3
4	6	7

Data in Character Arrays

Sometimes characters represent data that does not correspond to text, such as a DNA sequence. You can store this type of data in a character array, which has data type `char`. Character arrays use single quotes.

```
seq = 'GCTAGAATCC';  
whos seq
```

Name	Size	Bytes	Class	Attributes
seq	1x10	20	char	

Each element of the array contains a single character.

```
seq(4)
```

```
ans =  
    'A'
```

Concatenate character arrays with square brackets, just as you concatenate numeric arrays.

```
seq2 = [seq 'ATTAGAAACC']
```

```
seq2 =  
'GCTAGAATCCATTAGAAACC'
```

Character arrays are common in programs that were written before the introduction of double quotes for string creation in R2017a. All MATLAB functions that accept string data also accept char data, and vice versa.

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